Fiscal Year:	FY 2025	**Task Last Updated:**	FY 02/20/2025
PI Name:	Grabham, Peter Ph.D.		
Project Title:	A Determination of Bioactive Proteins Secreted by the Human Vasculature in Response to Low Dose Space Radiation. A Completion of Unfinished Research.		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	pwg2@cumc.columbia.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	646-761-1275
Organization Name:	Columbia University		
PI Address 1:	Center for Radiological Research		
PI Address 2:	630W W 168th St, VC 11-243		
PI Web Page:			
City:	New York	**State:**	NY
Zip Code:	10032-3702	**Congressional District:**	13
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	UNSOL-JSC
Start Date:	03/14/2022	**End Date:**	03/14/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/18/2024 per S. Mack-Phillips/JSC (Ed., 10/15/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Brown, Lewis M. Ph.D. (Quantitative Proteomics Center, Columbia University)		
Grant/Contract No.:	80NSSC22K0937		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Investigate the proteomics profile of human 3D microvessel models after exposure to the simplified Galactic Cosmic Radiation simulation (sim GCR sim) on the collapse of mature human microvessel models and the inhibition of developing microvessels (angiogenesis).		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:			

Task Progress:	Note per PI: "This work provided a clear demonstration that tools have been developed to identify and quantify bioactive proteins secreted by the human vasculature in response to low dose space radiation. The tools and approaches outlined here will be valuable in follow-on work." [Ed., 2/20/25]
Bibliography Type:	Description: (Last Updated: 03/04/2024)